

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

2023

Bachelor in Computer Applications

Course Title: **OOP in Java**

Code No: **CACS204** Semester: **III**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

2. What is java Buzzwords? Write a java program to find simple interest. Use command line argument to take input.
3. What are uses of super keyword? Write a java program to create base class Fruit which has name, taste and size as its attributes and method called eat() which describe name and its taste. Inherit the same in two other class Apple and Orange and override the eat() method to represent each fruit taste.
4. What is difference between String and StringBuffer class? Write a java program to identify the input string is palindrome or not?
5. Why we need file handling in java? Write java program to read file into a variable and then write a variable's content into another file.
6. What is the purpose of valueOf() method in Wrapper classes? Write a java program to generate random integer, double and bytes.
7. What is internal frames? Write a java program that display two internal frame within some parent frame.
8. What is JDBC? Write java program to connect database College and display all student information (Roll, Name, Address and Program) from Student table.

Group C

Attempt any TWO questions[2x10=20]

9. What is Constructor overloading? Write a class Distance containing private variable feet of type int and inches of type int, suitable constructors and three methods addDistance, subtractDistance and displayDistance for adding, subtracting and displaying objects. Write a separate class MyDistance containing main method to create, add, subtract and display distance objects.
10. a) What is difference between checked and unchecked exception? Write a java program that will read college name from keyboard and display it on screen. The program should throw an exception when length of college name is more than 50. b) What are methods used for inter-thread communication? Write a java

program to create two threads so that one thread prints even number and other thread prints odd number between 100 and 200.

- 11.** What is MVC design pattern? Write a GUI program using swing components to calculate sum and difference of two numbers. Use two text fields for input and pre-built dialog box for output. Your program should display sum if Add button and difference if subtract button is clicked.